OAK SOE OO

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SUMMARY

Multidisciplinary engineer studying Computer Engineering & Business undergraduate at HKUST. Worked across hardware, software, production systems, backtesting pipelines, and machine learning.

EDUCATION

Hong Kong University of Science and Technology (HKUST) Sep 2025 - Jun 2029 (expected)

Dual Degree Program in Technology and Management: BEng in Computer Engineering and BBA (undeclared)

- Merit-based full-ride Entrance Admission Scholarship (HK$1,225,000).
- HKSAR Government Belt and Road Scholarship (HK$925,000).

GCE International A-Levels (IAL) - *Self-studied* 2023 - 2025

- Earned 5 A* and 1 A, a result achieved by fewer than 0.01% (calculated) of candidates worldwide.
- Completed 5 A-Levels in 8 months.

EXPERIENCE

HKUST School of Humanities - *Research Assistant* Feb 2026 - Present

- Developing NLP pipelines and custom LLM frameworks to automate the analysis of large-scale qualitative datasets, streamlining the extraction of thematic insights from historical and social archives.
- Developing end-to-end data analytics workflows to transform unstructured humanities research into quantitative visualizations, enabling multidisciplinary teams to validate hypotheses through statistical modeling.

International Genetically Engineered Machine (iGEM), HKUST - *Dry Lab Team Lead* Feb 2026 - Present

- Lead a 9-member dry-lab team; apply machine learning to genetic engineering by training and fine-tuning transformer models, foundational models and CNNs to mine a non-model bacterium genome with data constraints.
- Selected from 100+ competitors; ran accelerated deep learning workloads on Linux GPU nodes using CUDA and PyTorch; presented the project at the Grand Jamboree in Paris 2026.

Rivermap Company Limited (Hong Kong) - *Research Analyst Intern* Jun 2026 - Sep 2026

- Design systems and database architecture, and build and operate an automated Python pipeline taking in large volumes of data from data vendors, normalise and group them to create smart-beta index strategies.
- Developed and deployed a full-stack internal management system with a React frontend, FastAPI backend, and relational database, replacing spreadsheet workflows; used daily by 18 staff across the firm.
- Built a centralised authentication and authorization system for use across the firm's 4+ internal and external applications.

Shun Lei Shwe Yi Co., Ltd. (Myanmar) - *Engineering Intern (Part-time)* Jan 2022 - Jun 2025

- Built and maintained document and data-management systems for 3 construction projects.

PROJECTS

ExposAI - *24-hour hackathon (team project)* Mar 2026

- Developed a dual-layer verification system combining cryptographic hashing on the Abelian (QDay) Blockchain with multimodal AI forensics via AWS Bedrock to differentiate between authentic and synthetic media
- Implemented a smart contract on the QDay Testnet to create a permanent, tamper-proof registry mapping unique image hashes to on-chain certificates.

COPS - *IoT water-filtration prototype* Oct 2025 - Jan 2026

- Built an Arduino-based prototype using IoT sensors and actuators that adjust filtration to measured water conditions.
- Won 2nd Prize at the Global Sustainability Challenge Regional Finals; presented at Hong Kong Techathon+.

AWARDS & COMPETITIONS

- Highest score in Myanmar, International Youth Maths Olympiad, Dec 2021.
- Gold Medal, Online Myanmar Mathematics Olympiad, Dec 2021.

SKILLS

PROGRAMMING	Python, C++, SQL, TypeScript, JavaScript
FRAMEWORKS & WEB	React, React Native, Node.js, FastAPI, Flask, HTML/CSS
DATA & ML	NumPy, NLP, LLM APIs, Prompt Design, deep learning (transformers, CNNs), Web Scraping
TOOLS & CLOUD	Git, Docker, CI/CD, AWS (incl. Bedrock), PostgreSQL, MySQL, Document Databases, Cybersecuirty
LANGUAGES	English~(fluent), Burmese~(native), Japanese~(conversational), Korean~(conversational)